

Learning to Route Under Load: An Inductive Graph Network that Amortizes Congestion-Aware Traffic Engineering for LEO Mega-Constellations

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Abstract—Low Earth orbit (LEO) mega-constellations carry traffic over a dense mesh of inter-satellite links whose topology, although time varying, is known in advance from orbital ephemeris. Under load, routing on propagation delay alone concentrates traffic on a few bottleneck links while alternative paths sit idle, wasting a large fraction of achievable performance. The congestion-aware optimum, the user equilibrium, removes this waste but requires the full demand matrix and an iterative solve at every slot, which does not scale to constellations of more than a thousand satellites with rapidly changing topology. We formalize routing under load on LEO as the amortization of a traffic-assignment equilibrium, decompose it into a price-prediction sub-problem, a decoding sub-problem, an inductive-transfer sub-problem, and a proactive sub-problem, and solve the first with an inductive graph neural network that predicts the equilibrium congestion price on each inter-satellite link in a single forward pass from cheap, observable load features. On a constellation simulator with a standard Bureau of Public Roads cost model, the network matches the user equilibrium in distribution while using one to two orders of magnitude less computation, recovers about four fifths of the gain when transferred zero shot to a shell twice as large, and, fed the predicted near future demand, sustains low travel time under drifting traffic hotspots where a reactive one slot lagged controller degrades below congestion-blind shortest path. We also report a negative result: a quantum-walk propagation operator that motivated this study offers no benefit once load features are available, and a plain graph convolution is preferable. We release the full pipeline and the honest record of the design search.

Index Terms—LEO satellite networks, non-terrestrial networks, routing, traffic engineering, graph neural networks, congestion control, user equilibrium, inductive learning.

I. INTRODUCTION

LOW Earth orbit (LEO) mega-constellations such as Starlink and Kuiper interconnect their satellites with laser inter-satellite links (ISLs) arranged in a regular grid, forming a wide-area packet network that moves with the constellation. Two features distinguish this network from a terrestrial one. First, the topology is time varying but *deterministic*: the position of every satellite, hence every ISL and its propagation delay, is known in advance from the orbital parameters. Second, the network is large and fast: a single shell contains a thousand or more satellites, and cross-plane links open and close as satellites traverse the constellation.

Routing such a network on propagation delay alone is attractive because the shortest path is cheap and, when lightly

loaded, near optimal. Under load it is wasteful: it funnels many flows onto the same minimum-hop links, creating hotspots, while geometrically longer but idle paths are ignored. We measure this waste directly (Section VII-A) and find that congestion-aware routing cuts total travel time by half or more once the network is moderately loaded.

The congestion-aware optimum is classical. Routing every flow so that none can unilaterally lower its cost yields the *user equilibrium* (UE), computed by iterating shortest-path assignment against a load-dependent cost. UE is the right target but expensive: it needs the entire demand matrix and many shortest-path rounds per slot. At constellation scale, with topology changing every few seconds, recomputing UE from scratch each slot is impractical.

This tension, a cheap-but-blind heuristic versus an expensive-but-optimal solve, motivates our approach: *amortize* the equilibrium. A graph neural network (GNN) observes cheap features (per-node traffic intensity and the load that one congestion-blind pass would induce) and predicts, in a single forward pass, the equilibrium congestion price on every ISL. Routing on the predicted cost recovers most of the congestion-aware gain at a fraction of the cost of solving UE. Because the model's parameters do not depend on the number of satellites, it is *inductive*: trained on a small shell, it applies unchanged to a larger one. And because the demand pattern drifts predictably (the dense-traffic band rotates under the constellation), feeding the predicted near future demand lets the router act *proactively*, before congestion forms. Figure 1 summarizes the pipeline.

Contributions.

- We formalize routing under load on LEO as the amortization of a traffic-assignment equilibrium and *decompose* it into four sub-problems (Section IV), each mapped to a measurable claim.
- **C1 (amortization)**. A one-shot GNN matches the user equilibrium in distribution (total travel time within two percent) at 20 to 27 $\times$ less computation than the iterative solve.
- **C2 (inductive scale generalization)**. Trained on small shells, the same model transferred zero shot to a shell twice as large is the best *deployable* policy, recovering about 79% of the blind-to-UE gain and beating every cheap baseline.
- **C3 (cost)**. We give a complexity analysis and wall-clock measurements showing the inference saving against UE is the operative reason to learn rather than solve.

TABLE I
NOTATION.

symbol	meaning
$G_t = (V, E_t)$	constellation graph at slot t
prop_{uv}	propagation delay of ISL (u, v)
cap	ISL capacity
$\mathcal{D} = \{(s, d, r)\}$	demand set (source, destination, rate)
x_{uv}	load (aggregate rate) on ISL (u, v)
$\text{cost}_{uv}(x)$	realized link cost under load (BPR)
g_{uv}^*	user-equilibrium congestion price of (u, v)
$\hat{g}_{uv}$	price predicted by the GNN
$h_v^{(l)}$	embedding of node v at layer l
$\text{TTT}(\pi)$	total travel time of routing policy π

- **C4 (proactive).** Under drifting hotspots, routing on predicted demand tracks the clairvoyant optimum, whereas a reactive one-slot-lagged controller degrades below congestion-blind shortest path as the drift grows.
- **Honest negative result.** The study began as a quantum-walk GNN; we report the full kill-gate record and why a plain graph convolution is the right operator here.

II. RELATED WORK

LEO routing and topology. The deterministic, time-varying structure of LEO ISL networks has been studied for low-latency routing and topology design [1], [2]. Packet-level simulators such as Hypatia [3] and StarPerf [4] reconstruct the moving topology from constellation parameters. These works route mostly on delay; we target the loaded regime where congestion dominates.

GNNs for networking. Graph convolutions [5], attention [6], and inductive sampling [7] underpin learning on graphs. RouteNet predicts per-path delay and jitter for network modeling [8]; reinforcement learning with GNNs has been applied to routing optimization [9]. Our model is supervised and predicts a congestion-price field decoded by shortest path, keeping inference to one forward pass and making inductivity straightforward.

Traffic engineering and equilibria. Load-dependent link cost and the user equilibrium are classical in transportation [10], [11], [12]; learning to route to reduce congestion has been studied for terrestrial networks [13], [14]. We bring this to LEO, where the value lies in amortizing the per-slot solve across a fast, scale-varying topology and in exploiting the predictable demand drift.

Quantum-walk networks. Continuous-time quantum walks have been used as a graph propagation operator [15]; we evaluate one and report it as a negative ablation.

III. SYSTEM MODEL

Table I lists the notation.

A. Time-varying constellation graph

A Walker shell of S satellites in P planes is propagated on circular orbits. Each satellite holds two intra-plane ISLs and up to two cross-plane ISLs, forming a +Grid (Fig. 3). The topology over a horizon is a known sequence $\{G_t\}_{t=0}^{T-1}$,

$G_t = (V, E_t)$, with $|V| = S$. Each ISL $(u, v) \in E_t$ has a propagation delay $\text{prop}_{uv} = \|p_u - p_v\|/c$ from the known positions p , and a capacity cap .

B. Demand, flows, and link load

A demand set $\mathcal{D} = \{(s_k, d_k, r_k)\}$ requests rate r_k from s_k to d_k . We draw demands from a gravity model whose hotspots concentrate near a longitude band ϕ_t that rotates between slots, a proxy for the rotating dense-population and sub-solar region; the per-slot drift $\Delta = \phi_{t+1} - \phi_t$ is known. A routing policy maps each demand k to a path P_k in G_t . Writing $f_k \geq 0$ for the rate on P_k (here $f_k = r_k$ on a single path), the load on an ISL is

$$x_{uv} = \sum_k r_k \mathbb{1}[(u, v) \in P_k], \quad (1)$$

subject to flow conservation at every node $v \notin \{s_k, d_k\}$:

$$\sum_{u:(u,v) \in E_t} f_{uv}^k = \sum_{w:(v,w) \in E_t} f_{vw}^k. \quad (2)$$

C. Congestion cost and realized travel time

The realized cost of a link under load follows the Bureau of Public Roads (BPR) form [11],

$$\text{cost}_{uv}(x) = \text{prop}_{uv} \left(1 + \alpha (x_{uv}/\text{cap})^\beta \right), \quad \alpha = 0.15, \beta = 4, \quad (3)$$

which is smooth, strictly increasing, and convex in x . The total travel time of a policy π inducing loads $x(\pi)$ is

$$\text{TTT}(\pi) = \sum_k r_k \sum_{(u,v) \in P_k} \text{cost}_{uv}(x_{uv}(\pi)). \quad (4)$$

D. User equilibrium and system optimum

Two operating points bound any router. The *user equilibrium* (UE) satisfies Wardrop's first principle: for every demand, all used paths have equal and minimal cost. Equivalently, UE is the unique minimizer of the Beckmann potential [12]

$$x^{\text{UE}} = \arg \min_{x \in \mathcal{F}} \sum_{(u,v) \in E_t} \int_0^{x_{uv}} \text{cost}_{uv}(w) dw, \quad (5)$$

over the polytope $\mathcal{F}$ of loads induced by feasible flows (1)–(2); with (3) the integral is $\text{prop}_{uv}(x + \frac{\alpha \text{cap}}{\beta+1}(x/\text{cap})^{\beta+1})$. The *system optimum* (SO) instead minimizes (4) directly, $x^{\text{SO}} = \arg \min_{x \in \mathcal{F}} \text{TTT}(x)$, and lower-bounds UE; their ratio is the price of anarchy. We use UE as the deployable congestion-aware reference and note SO as a tighter (more expensive) ceiling.

We compute UE by the method of successive averages (MSA): starting from an all-or-nothing assignment y^0 on free-flow cost, iterate

$$x^{k+1} = x^k + \frac{1}{k+1} (y^k - x^k), \quad y^k = \text{AON}(\text{cost}(x^k)), \quad (6)$$

where AON routes every demand on the shortest path under the current cost. Each iteration is $|\mathcal{D}|$ shortest paths; convergence needs T iterations.

E. The amortization problem

Blind shortest path is one AON pass and is congestion blind; UE is T passes and is congestion optimal but does not scale. We seek a parametric map f_θ from cheap, observable state $\mathcal{S}_t$ to a routing π_θ that approaches UE in one forward pass and whose parameters θ are independent of S :

$$\min_{\theta} \mathbb{E}_{\mathcal{D}, G} [\text{TTT}(\pi_\theta)] \quad \text{s.t.} \quad \text{cost}(\pi_\theta) \ll \text{cost}(\text{UE}). \quad (7)$$

IV. PROBLEM DECOMPOSITION

We decompose (7) into four sub-problems (Fig. 2), each mapped to a claim.

(SP1) Price prediction. Approximate the UE congestion price field

$$g_{uv}^* = \text{cost}_{uv}^{\text{UE}} / \text{prop}_{uv} - 1 \geq 0, \quad \hat{g} = f_\theta(\mathcal{S}_t), \quad (8)$$

so that routing on $\text{prop}(1 + \hat{g})$ reproduces x^{UE} . Maps to **C1**.

(SP2) Decoding. Turn a price field into a routing by one shortest-path assignment, $\pi = \text{AON}(\text{prop}(1 + \hat{g}))$. The decode is shared across all demands, so a single $\hat{g}$ routes the whole matrix.

(SP3) Inductive transfer. Require θ to be size invariant so that f_θ trained on $\{G^{\text{small}}\}$ applies to a larger G^{large} zero shot. Maps to **C2**. A message-passing f_θ with shared weights and a local readout satisfies this by construction (Section V-E).

(SP4) Proactivity. Under drift Δ , predict next-slot demand $\hat{\mathcal{D}}_t$ from $\mathcal{D}_{t-1}$ and the known Δ , and decode on $f_\theta(\mathcal{S}(\hat{\mathcal{D}}_t))$ so the route pre-empts the moving hotspot. Maps to **C4**.

Cost (SP2 plus inference) versus the UE solve gives **C3** (Section V-G).

V. METHOD

A. Observable features (input to SP1)

For a slot we form per-node features $\phi_v = [\text{out}_v, \text{in}_v, x_v^{0,\text{out}}, x_v^{0,\text{in}}]$, where $\text{out}_v, \text{in}_v$ are the rate originating and terminating at v , and $x_v^{0,\cdot}$ aggregate the *blind load* $x^0 = \text{AON}(\text{prop})$ leaving and entering v . Per-edge features are $[\text{prop}_{uv}, x_{uv}^0]$. The blind load is one AON pass (no iteration) and is the most informative input, since the equilibrium price of a link is largely a function of how overloaded that link would be under naive routing. All features are scaled to be size invariant.

B. Edge-price GNN (model for SP1)

Node embeddings are initialized $h_v^{(0)} = \text{ReLU}(W_{\text{in}}\phi_v)$ and updated for $l = 0, \dots, K-1$ by a symmetric-normalized graph convolution

$$h_v^{(l+1)} = \text{ReLU}\left(W^{(l)} \sum_u \hat{A}_{vu} h_u^{(l)}\right), \quad (9)$$

where $\hat{A} = \tilde{D}^{-1/2}(A+I)\tilde{D}^{-1/2}$ and $\tilde{D}$ is the degree of $A+I$. A per-edge readout produces the price

$$\hat{g}_{uv} = \text{softplus}\left(\text{MLP}([h_u, h_v, \text{prop}_{uv}, x_{uv}^0])\right). \quad (10)$$

Algorithm 1 One-slot congestion-aware routing

Require: graph G_t , demand $\mathcal{D}$ (or forecast $\hat{\mathcal{D}}$), weights θ

- 1: $x^0 \leftarrow \text{AON}(\text{prop})$ $\triangleright$ one blind pass = features
- 2: $\phi \leftarrow \text{FEATURES}(\mathcal{D}, x^0)$
- 3: $\hat{g} \leftarrow f_\theta(\phi, G_t)$ $\triangleright$ one GNN forward, Eq. (10)
- 4: $\pi \leftarrow \text{AON}(\text{prop}(1 + \hat{g}))$ $\triangleright$ one decode pass
- 5: **return** π

C. Training objective

We regress the log-price over a dataset $\mathcal{I}$ of (shell, demand) instances:

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \frac{1}{|E_i|} \sum_{(u,v) \in E_i} \left(\log(1 + \hat{g}_{uv}) - \log(1 + g_{uv}^*) \right)^2. \quad (11)$$

The $\log(1 + \cdot)$ scale tames the heavy tail of g^* at saturated links. Targets g^* come from solving UE (6) on the training instances offline.

D. Decoding (SP2)

Given $\hat{g}$, the routing is a single shortest-path assignment $\pi = \text{AON}(\text{prop}(1 + \hat{g}))$, shared across the whole demand matrix. Algorithm 1 states inference.

E. Inductive transfer (SP3)

The parameters $\theta = \{W_{\text{in}}, W^{(l)}, \text{MLP}\}$ have dimensions set by the feature width and hidden size only, not by S or $|E|$. The aggregation (9) and readout (10) are defined pointwise over nodes and edges. Hence f_θ evaluates on any graph, and a model trained on small shells applies to a larger shell with no change, which is exactly SP3.

F. Proactive variant (SP4)

Under known drift Δ , the next-slot hotspot band is $\phi_t = \phi_{t-1} + \Delta$, so the demand forecast $\hat{\mathcal{D}}_t$ is available. The proactive router decodes on $f_\theta(\mathcal{S}(\hat{\mathcal{D}}_t))$; the reactive baseline decodes on $f_\theta(\mathcal{S}(\mathcal{D}_{t-1}))$ and is therefore one step behind.

G. Complexity (C3)

A forward pass costs $O(K|E|d + K|V|d^2 + |E|d^2)$ for hidden width d ; a decode pass costs $O(|\mathcal{D}|(|E| + |V| \log |V|))$. The full router is one feature pass, one forward, and one decode, i.e. two AON passes plus the forward. UE by MSA (6) costs T AON passes, $T \approx 20$ in our runs. The predicted speedup is thus $\approx T/2$ to T , matching the measured 19 to $27\times$ (Table III).

VI. EXPERIMENTAL SETUP

We simulate Walker shells with a pure-kinematic propagator (no packet-level dependency, fully reproducible from parameters): an Iridium-like shell (66 satellites, diameter 9) and Starlink-like 53° shells of 132 ($d=16$), 264 ($d=28$), and 1584 satellites. Capacity is uniform; demands follow the drifting gravity model. We report total travel time (4) relative to blind shortest path (lower is better; blind = 1.00), averaged over

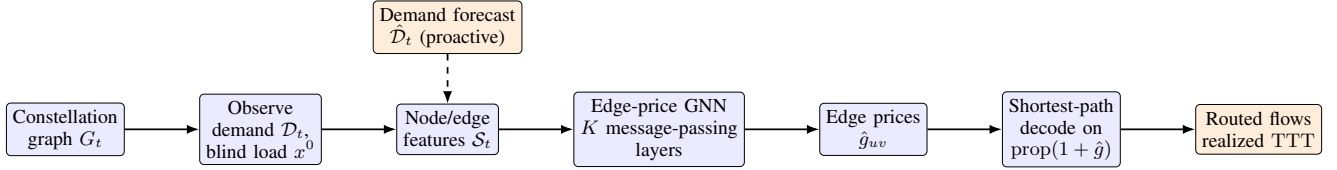


Fig. 1. End-to-end pipeline. Cheap observable state (demand and the load that one congestion-blind pass would induce) feeds an edge-price GNN that predicts the equilibrium congestion price per ISL in one forward pass; a single shortest-path decode then routes the whole demand matrix. The proactive variant (dashed) feeds the predicted next-slot demand.

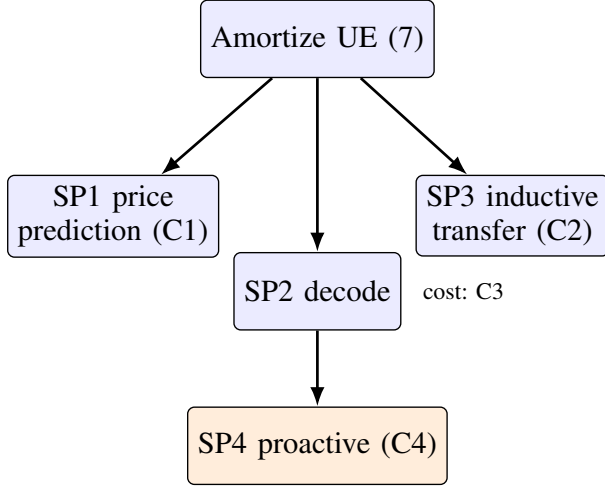


Fig. 2. Decomposition of the amortization problem into four sub-problems, each mapped to a measured claim.

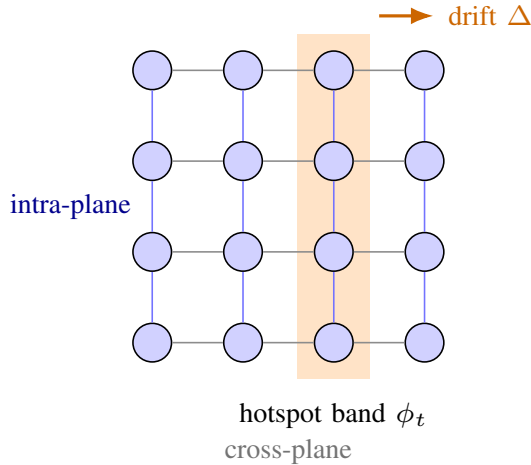


Fig. 3. LEO + Grid ISL topology (torus). Intra-plane links (vertical) are stable; cross-plane links (horizontal) vary with the geometry. The shaded band is the traffic hotspot region, which drifts by a known Δ per slot.

multiple demand instances and seeds with standard deviations. Baselines: *blind*; *geographic* greedy (great-circle next hop, with a shortest-path fallback so all flows are delivered); *one-step* congestion correction (route on one blind-load pass); *GNN* (ours); and *UE* (the congestion-aware reference). Learned results are param-matched across operators and reported with error bars; metrics are survivorship free.

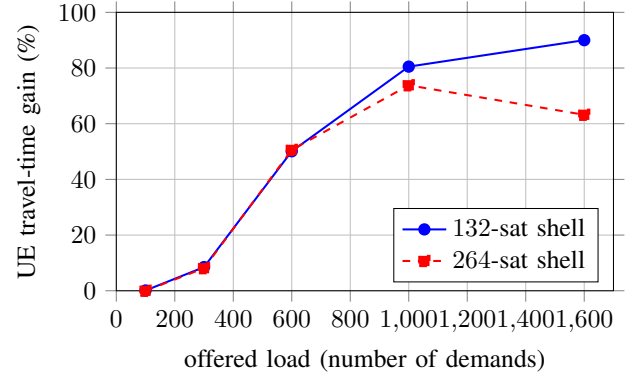


Fig. 4. Congestion headroom: the gain of the user equilibrium over congestion-blind shortest path grows with load. None at light load; half or more under load.

TABLE II

TOTAL TRAVEL TIME RELATIVE TO BLIND (LOWER IS BETTER; BLIND = 1.00). MEAN OVER INSTANCES AND SEEDS. UE IS THE EXPENSIVE REFERENCE; THE GNN IS THE DEPLOYABLE POLICY. SO ROW IS A PLACEHOLDER PENDING THE FRANK-WOLFE RUN.

split	UE (ref.)	geographic	one-step	GNN	blind
in-distribution	0.46	0.78	1.04	0.47	1.00
OOD (2× shell)	0.23	0.80	0.55	0.39	1.00
SO (projected)	<i>TBD</i>	–	–	–	–

VII. RESULTS

A. Headroom for congestion awareness

Figure 4 sweeps offered load. At light load, blind shortest path is near optimal and UE offers nothing. As load grows, blind concentrates traffic on bottleneck links (its peak utilization reaches several times capacity while alternates idle) and UE recovers 8, 50, 80, and 90% of travel time. The congestion task, not static routing, is where a learned router helps.

B. The GNN matches UE cheaply and transfers (C1, C2, C3)

Figure 5 and Table II report TTT relative to blind. In distribution, the GNN (0.47) matches UE (0.46). Transferred zero shot to a shell twice as large, the GNN (0.39) is the best *deployable* policy, recovering about 79% of the blind-to-UE gain and beating every cheap baseline: one-step correction overshoots (worse than blind in distribution), and geographic greedy stalls on the ISL grid, requiring fallback for 53 to 68% of flows. The GNN reaches this at 20 to 27× less computation than MSA (Table III).

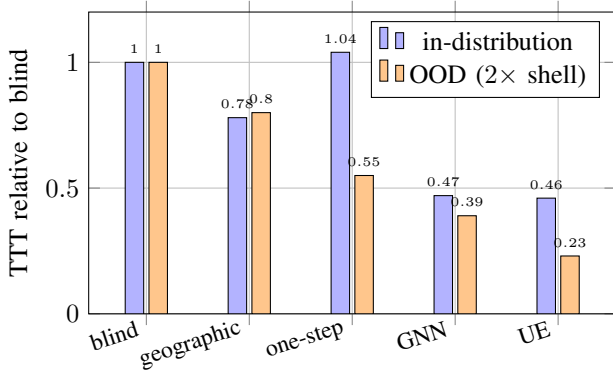


Fig. 5. Travel time relative to blind (lower is better). The GNN matches UE in distribution and is the best deployable policy zero shot; the cheap heuristics fail.

TABLE III

INFERENCE COST: ONE GNN FORWARD PLUS ROUTING VERSUS THE MSA SOLVE FOR UE (WALL-CLOCK, CPU). THE 1584-SATELLITE ROW IS A PLACEHOLDER PENDING THE SERVER RUN.

shell	sats	GNN (s)	UE/MSA (s)	speedup
132-sat	132	0.10	2.77	27×
264-sat	264	0.71	13.7	19×
1584-sat	1584	TBD	TBD	(proj. ~20)

C. Proactive routing under drift (C4)

Figure 6 sweeps the per-slot hotspot drift. At zero drift the reactive and proactive routers are identical, as they must be. As the hotspot moves, the reactive lag-1 router routes for yesterday's load and degrades, passing blind at moderate drift and reaching $1.6\times$ blind at the largest drift. The proactive router, fed the predicted demand, tracks the clairvoyant equilibrium throughout; the advantage grows from 0 to 76%.

D. Operator ablation, including the quantum walk (negative)

Figure 7 compares three param-matched propagation operators: a local graph convolution (GCN), a global classical-diffusion operator (Heat), and a global quantum-walk operator (QW). Once the blind-load features are present, predicting the price is largely a local function of a link's own load, so the plain GCN is best and the global operators add variance without benefit. The quantum-walk operator, which originally motivated this work, is *not* preferred; we keep it as an honest negative ablation.

VIII. DISCUSSION AND LIMITATIONS

The simulator is kinematic and flow level, not packet level; it captures geometry-driven delay and load-induced congestion through the BPR model, the mechanism the method targets, but does not model queueing dynamics or protocol behavior. UE is a selfish equilibrium and not the system optimum, which is why some policies record travel time slightly below UE; a system-optimal reference (placeholder in Table II) would tighten the upper bound. The zero-shot transfer leaves a residual gap to UE on the larger shell, driven partly by one high-variance seed; training across a range of shell sizes

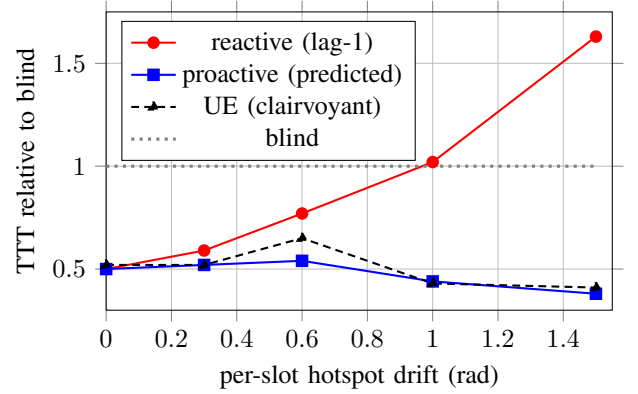


Fig. 6. Proactive versus reactive routing under drifting hotspots. The reactive lag-1 router degrades past blind as the hotspot moves; the proactive router tracks the clairvoyant optimum.

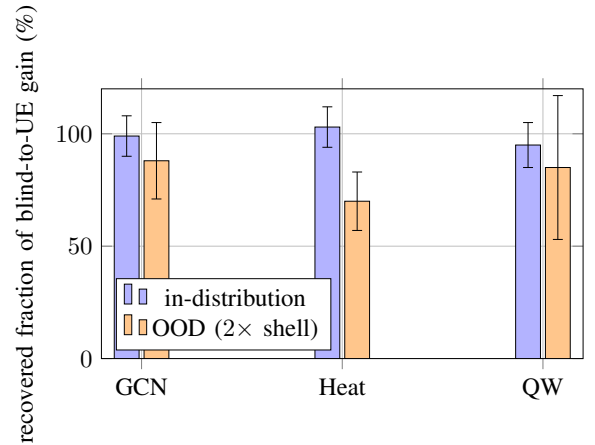


Fig. 7. Operator ablation (recovered fraction of the blind-to-UE gain, mean $\pm$ std). The plain GCN is best; the global quantum-walk operator does not help and is the most variable.

is a natural way to close it. The 1584-satellite validation (Tables II, III) is in progress. Finally, the proactive variant assumes the demand drift is predictable, which holds for the diurnal rotation of the dense-traffic band but not for sudden, unforecastable surges.

IX. CONCLUSION

We presented an inductive graph network that amortizes congestion-aware traffic engineering for LEO mega-constellations. After formalizing routing under load as the amortization of a traffic-assignment equilibrium and decomposing it into four sub-problems, we solve the price-prediction core with a GNN that predicts the user-equilibrium congestion price on every ISL in one forward pass from cheap load features. It matches the equilibrium in distribution at one to two orders of magnitude less computation, transfers zero shot to larger shells as the best deployable policy, and, fed predicted demand, sustains low travel time under drifting hotspots where reactive control fails. We also documented why an initially appealing quantum-walk operator did not earn its place, and released the full pipeline and design record so the negative results are reusable.

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